

Long-Run Economic Growth

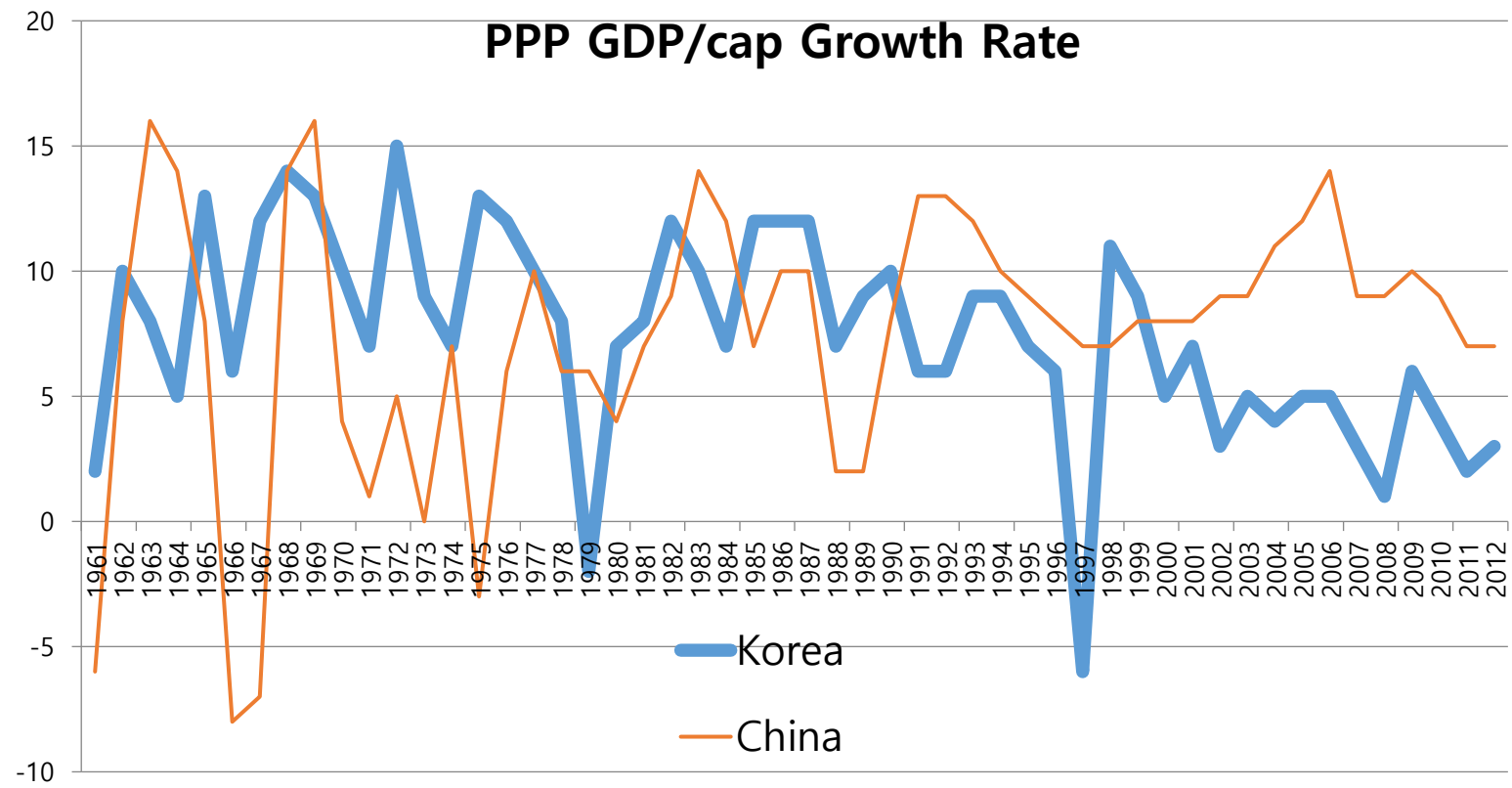
(Based on Chapter 13 of your text)

Relation Between Economic Growth and Adult Human Height Growth

A century of trends in adult human height Growth

<https://elifesciences.org/content/5/e13410>

- we found that the largest gains since the 1896 birth cohort occurred in South Korean women and Iranian men, who became 20.2 cm and 16.5 cm taller, respectively. As a result, South Korean women moved from the fifth shortest to the top tertile of tallest women in the world over the course of a century. Men in South Korea also had large gains relative to other countries, by 15.2 cm.



Comparing Economic Growth by Countries

- Nominal GDP vs. Real GDP per capita
- Real GDP per capita removes the effects from price changes, and population changes
- Average annual growth rate of GDP:

t=	0	1	2	.	.	n
	gdp_0	$gdp_0 * (1 + g)$	$gdp_0 * (1 + g)^2$			$gdp_0 * (1 + g)^n$
	So, $gdp_n = gdp_0 * (1 + g)^n$					
	$1 + g = \sqrt[n]{\frac{gdp_n}{gdp_0}}, \text{ So the growth rate (\%)} = (\sqrt[n]{\frac{gdp_n}{gdp_0}} - 1) * 100$					
Example:	gdp 2000		3000			
	gdp 2010		5000			$(\sqrt[10]{\frac{5000}{3000}} - 1) * 100(\%) = 5.2\%$
average annual growth rate =			5.2%			

Growth Rates : The **Rule of 70**

The **Rule of 70** tells us that the time it takes a variable that grows gradually over time to double is approximately 70 divided by that variable's annual growth rate.

$$\text{Number of years for variable to double} = \frac{70}{\text{Annual growth rate of variable}}$$

Ex) Annual growth rate: 2%

Years require for GDP doubled: $(70 / 2) = 35$ years

$$(1 + 0.02)^{35} = 1.9998895$$

The Sources of Long-Run Growth

- **Labor productivity**, often referred to simply as **productivity**, is output per worker.
- **Physical capital** consists of human-made resources such as buildings and machines.
- **Human capital** is the improvement in labor created by the education and knowledge embodied in the workforce.
- **Technology** is the technical means for the production of goods and services.

Accounting for Growth: The Aggregate Production Function

The **aggregate production function** is a hypothetical function that shows how productivity (real GDP per worker) depends on the quantities of physical capital per worker and human capital per worker as well as the state of technology.

$$\text{Aggregate production function: } Y/L = f(K/L, H/L, T)$$

Accounting for Growth: The Aggregate Production Function

Economists Barry Bosworth and Susan Collins of the Brookings Institution used the following aggregate production function:

$$\text{GDP per worker} = T \times (\text{Physical capital per worker})^{0.4} \times (\text{Human capital per worker})^{0.6}$$

- § Using this function, they tried to explain why China grew faster than India between 1978 and 2004.
- § About half the difference, they found, was due to China's higher levels of investment spending, which raised its level of physical capital per worker faster than India's.
- § The other half was due to faster Chinese technological progress.

Growth Accounting: An Example

Assumptions:

Output has grown 3% per year.

The labor force has grown 1% per year.

The quantity of physical capital has grown at 4% per year.

The average education level hasn't changed.

Each 1% increase in K/L -, other things equal, raises productivity (Y/L) by 0.3%.

a. How fast has productivity grown? : Productivity (Y/L) growth = $3\% - 1\% = 2\%$ (see below)

b. How fast has physical capital per worker grown? (K/L) growth = $4\% - 1\% = 3\%$

c. How much has growing physical capital per worker contributed to productivity growth?

$3\% \cdot 0.3\% = 0.9\%$ What percentage of productivity growth is that? $0.9\% / 2\% \cdot 100 = 45\%$

d. How much has technological progress contributed to productivity growth? $2\% - 0.9\% = 1.1\%$

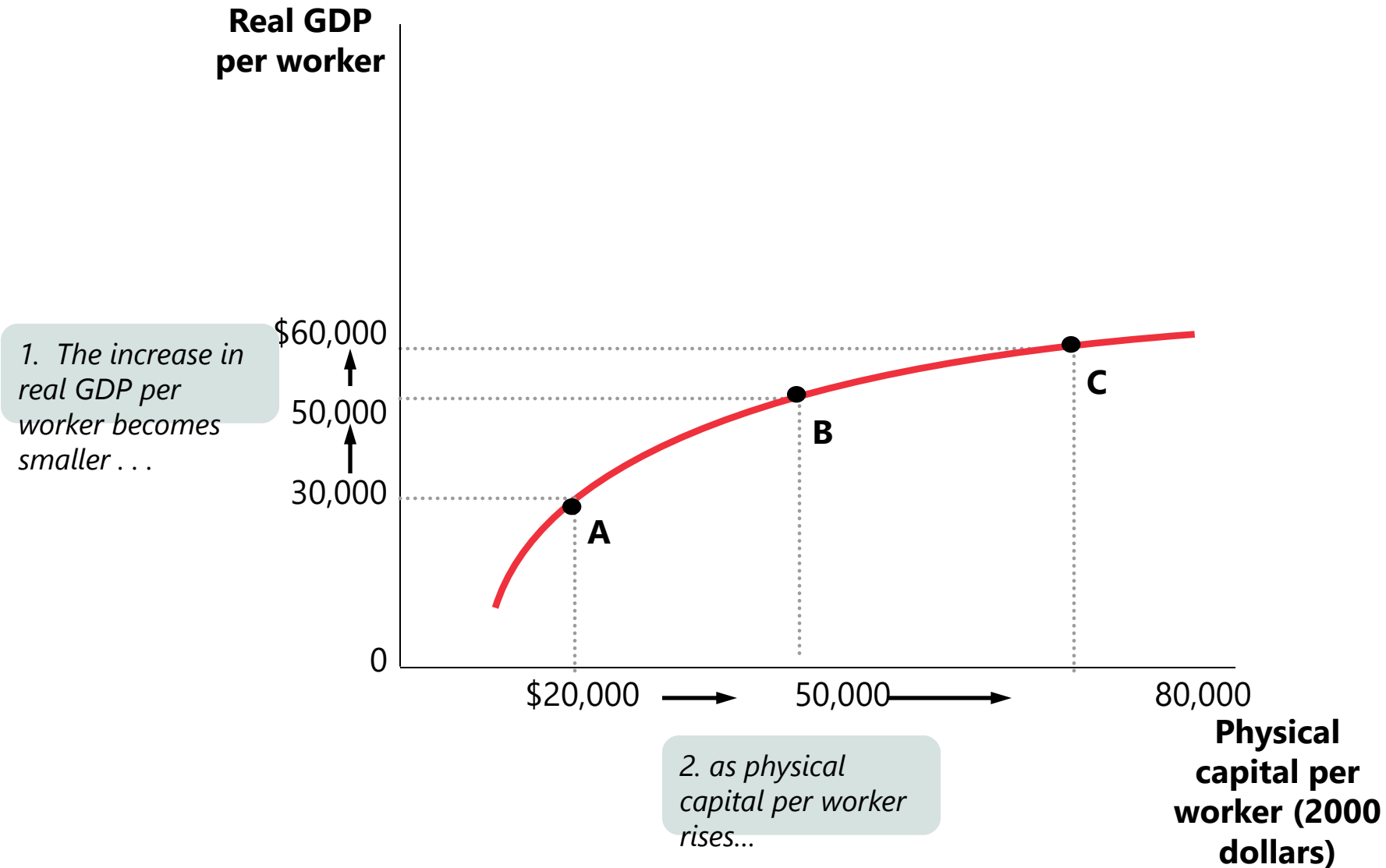
What percentage of productivity growth is that? $100\% - 45\% = 55\%$

* Growth rate of (Y/L) = (growth rate of Y) – (growth rate of L)

Diminishing Returns to Physical Capital

§ An aggregate production function exhibits **diminishing returns to physical capital** when, holding the amount of human capital and the state of technology fixed, each successive increase in the amount of physical capital leads to a smaller increase in productivity.

Physical Capital and Productivity



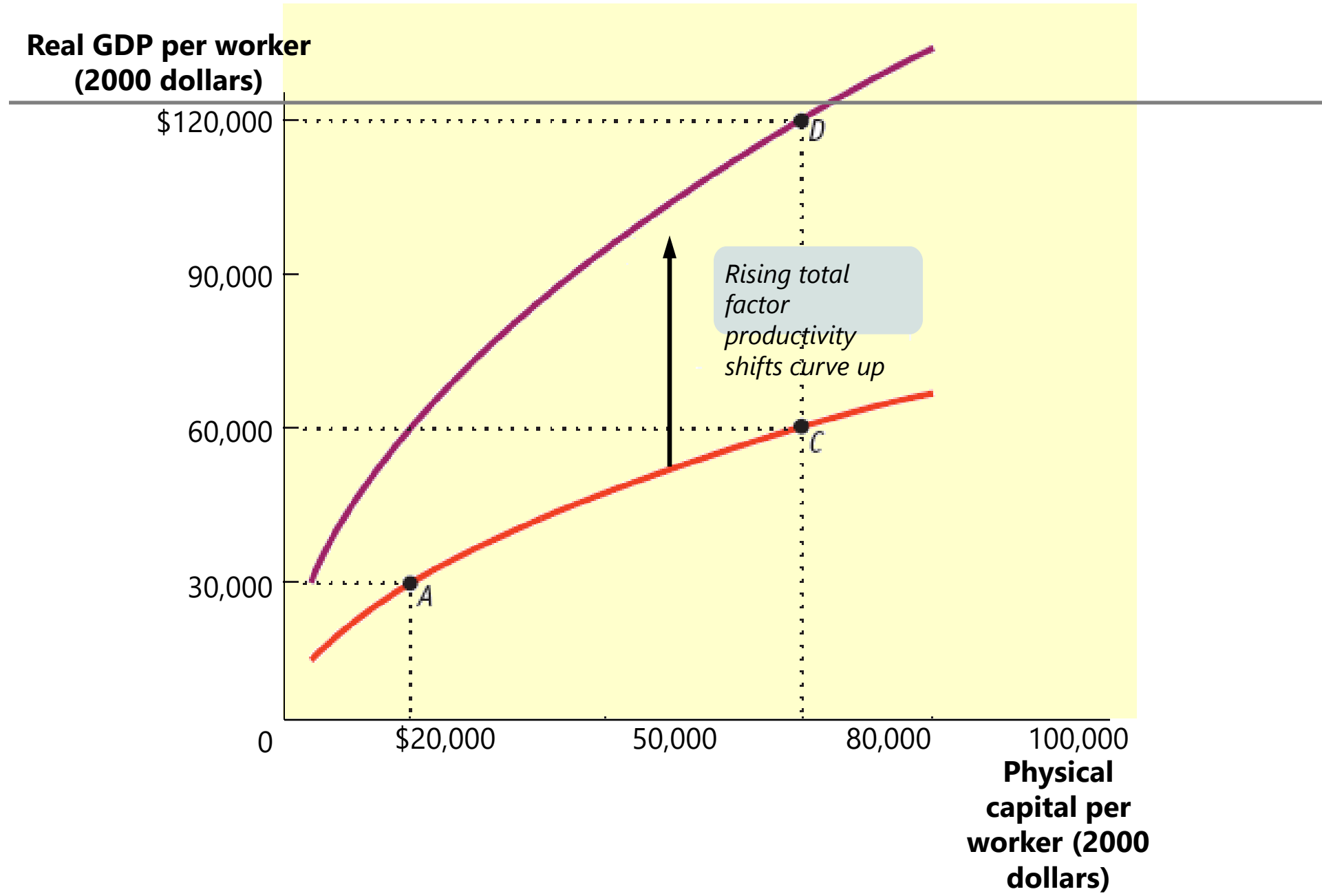
It May Be Diminished ... But It's Still Positive

- ***Diminishing returns to physical capital*** is an "***other things equal***" statement: holding the amount of human capital per worker and the technology fixed, each successive increase in the amount of physical capital per worker results in a smaller increase in real GDP per worker.
- This doesn't mean that real GDP per worker eventually falls as more and more physical capital is added. It's just that the *increase* in real GDP per worker gets smaller and smaller, albeit remaining at or above zero. So an increase in physical capital per worker will never reduce productivity.
- Due to diminishing returns, at some point increasing the amount of physical capital per worker no longer produces an economic payoff.

The Wal-Mart Effect

- After 20 years of being sluggish, U.S. productivity growth accelerated sharply (grew at a much faster rate) in the late 1990s. What caused that acceleration? Was it the rise of the Internet?
- According to McKinsey, the major source of productivity improvement after 1995 was a surge in output per worker in retailing—stores were selling much more merchandise per worker.
- Why? Wal-Mart has been a pioneer in using modern technology (for example, computers) to improve productivity.
- A lot of economic growth comes from everyday improvements rather than glamorous new technologies.

Technological Progress and Productivity Growth



What about Natural Resources?

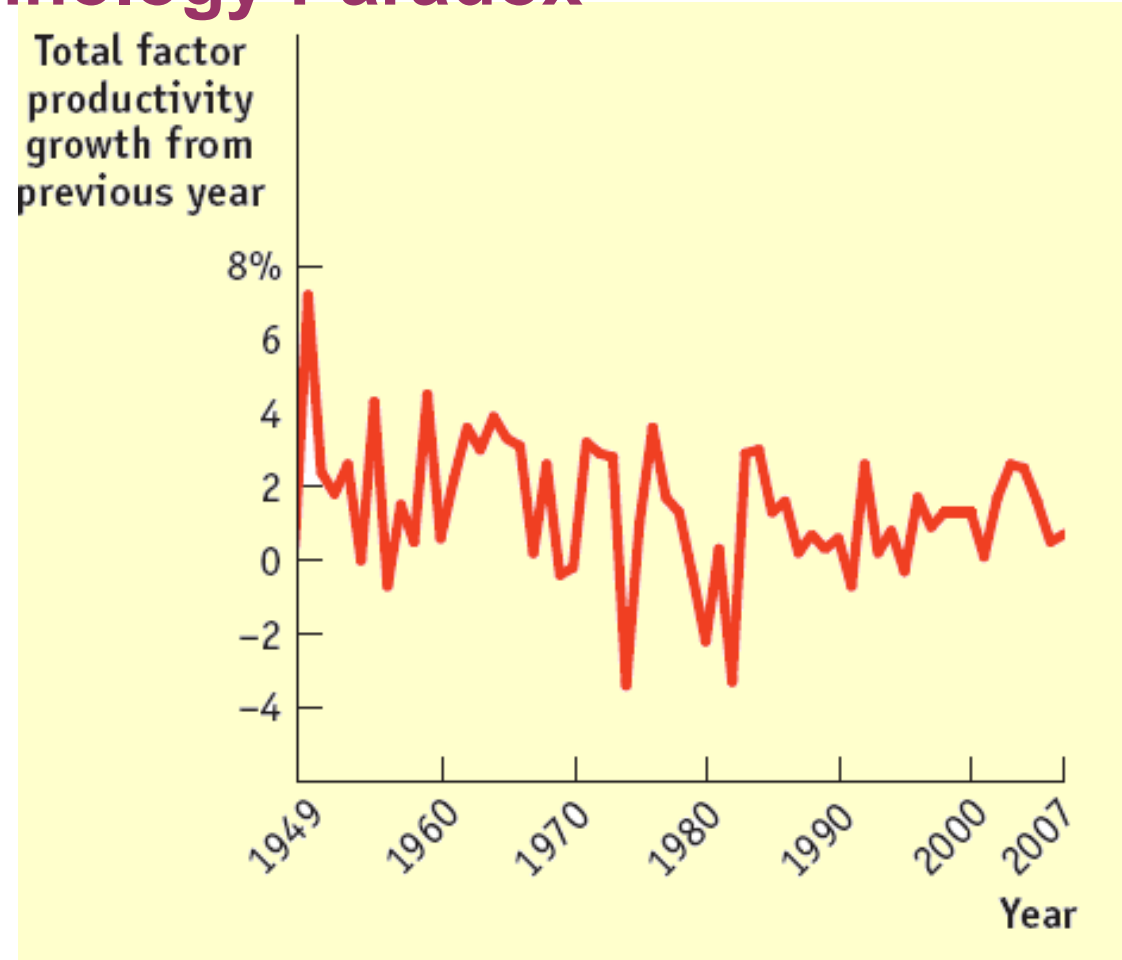
- In contrast to earlier times, in the modern world, natural resources are a much less important determinant of productivity than human or physical capital for the great majority of countries.
- For example, some nations with very high real GDP per capita, such as Japan, have very few natural resources. Some resource-rich nations, such as Nigeria (which has sizable oil deposits), are very poor.

The Information Technology Paradox

§ Many economists were puzzled by the slowdown in the U.S. growth rate of labor productivity—a fall from an average annual growth rate of 3% in the late 1960s to slightly less than 1% in the mid-1980s.

§ This was surprising given that there appeared to be rapid progress in technology.

§ *Why didn't information technology show large rewards?*



The Information Technology Paradox

- MIT economics professor and Nobel laureate Robert Solow, a pioneer in the analysis of economic growth, declared that the information technology revolution could be seen everywhere except in the economic statistics.
- Paul David suggested that a new technology doesn't yield its full potential if you use it in old ways.
- Productivity would take off when people really changed their way of doing business to take advantage of the new technology—such as, replacing letters and phone calls with electronic communications.
- Sure enough, productivity growth accelerated dramatically in the second half of the 1990...

Why Growth Rates Differ

- A number of factors influence differences among countries in their growth rates.
- These are government policies and institutions that alter:
 - savings and investment spending.
 - foreign investment.
 - education.
 - Infrastructure.
 - research and development.
 - political stability.
 - the protection of property rights.

Case Study – Vietnam's economic miracle

- <https://www.weforum.org/agenda/2018/09/how-vietnam-became-an-economic-miracle/>

Why Growth Rates Differ

Human Capital in Latin America and East Asia

	<u>Latin America</u>		<u>East Asia</u>	
	1960	2000	1960	2000
Percentage of population with no schooling	37.90%	14.60%	52.50%	19.80%
Percentage of population with high school or above	5.9	19.5	4.4	26.5

Inventing R&D

- Thomas Edison is best known as the inventor of the light bulb and the phonograph. But his biggest invention was “research and development”!!!
- In 1875 Edison created something new: his Menlo Park, New Jersey, laboratory employed 25 men full-time to generate new products and processes for business. (<http://www.edisonnj.org/menlopark/>)
- In other words, he did not set out to pursue a particular idea and then cash in. He created an organization whose purpose was to create new ideas year after year.
- **Research and development**, or **R&D**, is spending to create and implement new technologies.

The Role of Government in Promoting Economic Growth

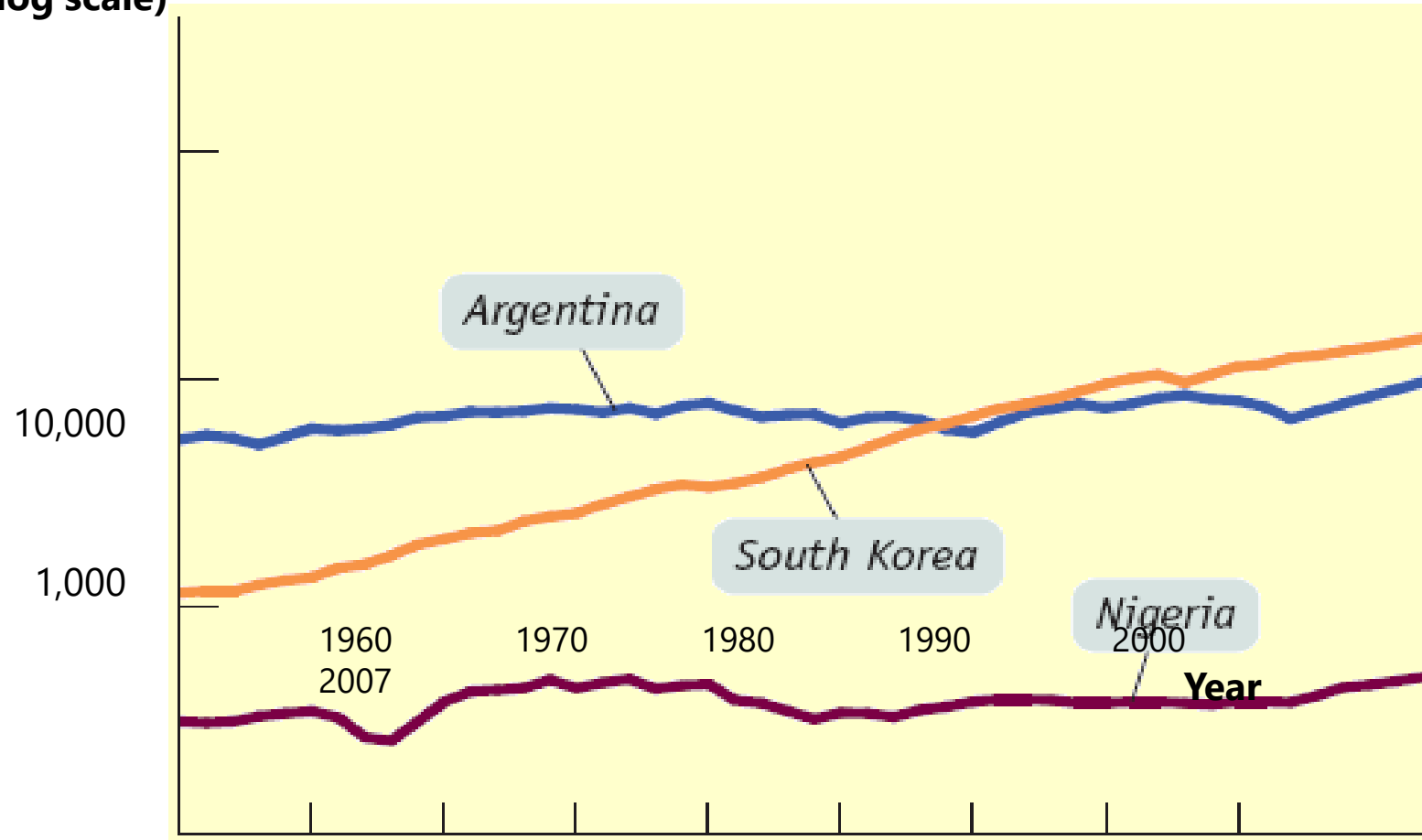
- Political stability and protection of property rights are crucial ingredients in long-run economic growth.
- Even when governments aren't corrupt, excessive government intervention can be a brake on economic growth.
- If large parts of the economy are supported by government subsidies, protected from imports, or otherwise insulated from competition, productivity tends to suffer because of a lack of incentives.

The Brazilian Breadbasket

- In recent years, Brazil's economy has made a strong showing, especially in agriculture.
- This success depends on exploiting a natural resource, the tropical savannah land known as the *cerrado*.
- A combination of three factors changed this land into a useable resource:
 - technological progress due to research and development
 - improved economic policies
 - addition of physical capital
- Brazil has already overtaken the United States as the world's largest beef exporter and may not be far behind in soybeans.

Success, Disappointment, and Failure

Real GDP per
capita (log scale)



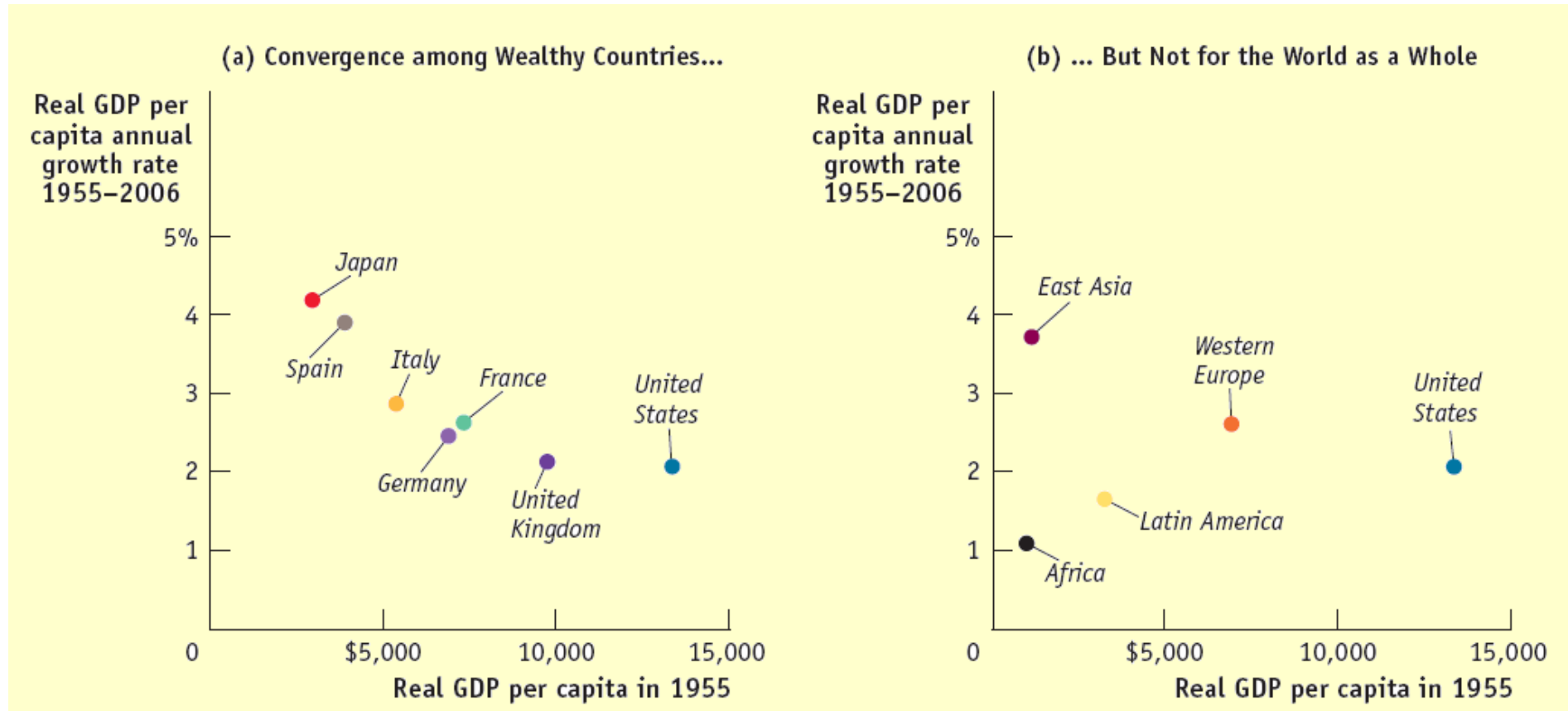
Success, Disappointment, and Failure

- The world economy contains examples of success and failure in the effort to achieve long-run economic growth.
- ***East Asian*** economies have done many things right and achieved very high growth rates.
- In ***Latin America***, where some important conditions are lacking, growth has generally been disappointing.
- In ***Africa***, real GDP per capita has declined for several decades, although there are some signs of progress now.

Success, Disappointment, and Failure

- The growth rates of economically advanced countries have converged, but not the growth rates of countries across the world.
- This has led economists to believe that the **convergence hypothesis** fits the data only when factors that affect growth, such as education, infrastructure, and favorable policies and institutions, are held equal across countries.

Are Economies Converging?



Success, Disappointment, and Failure

- **East Asia's** spectacular growth was generated by high savings and investment spending rates, emphasis on education, and adoption of technological advances from other countries.
- Poor education, political instability, and irresponsible government policies are major factors in the slow growth of **Latin America**.
- In **sub-Saharan Africa**, severe instability, war, and poor infrastructure—particularly affecting public health—have resulted in a catastrophic failure of growth. Encouragingly, the economic performance since the mid-1990s has been much better than in preceding years.

Is World Growth Sustainable?

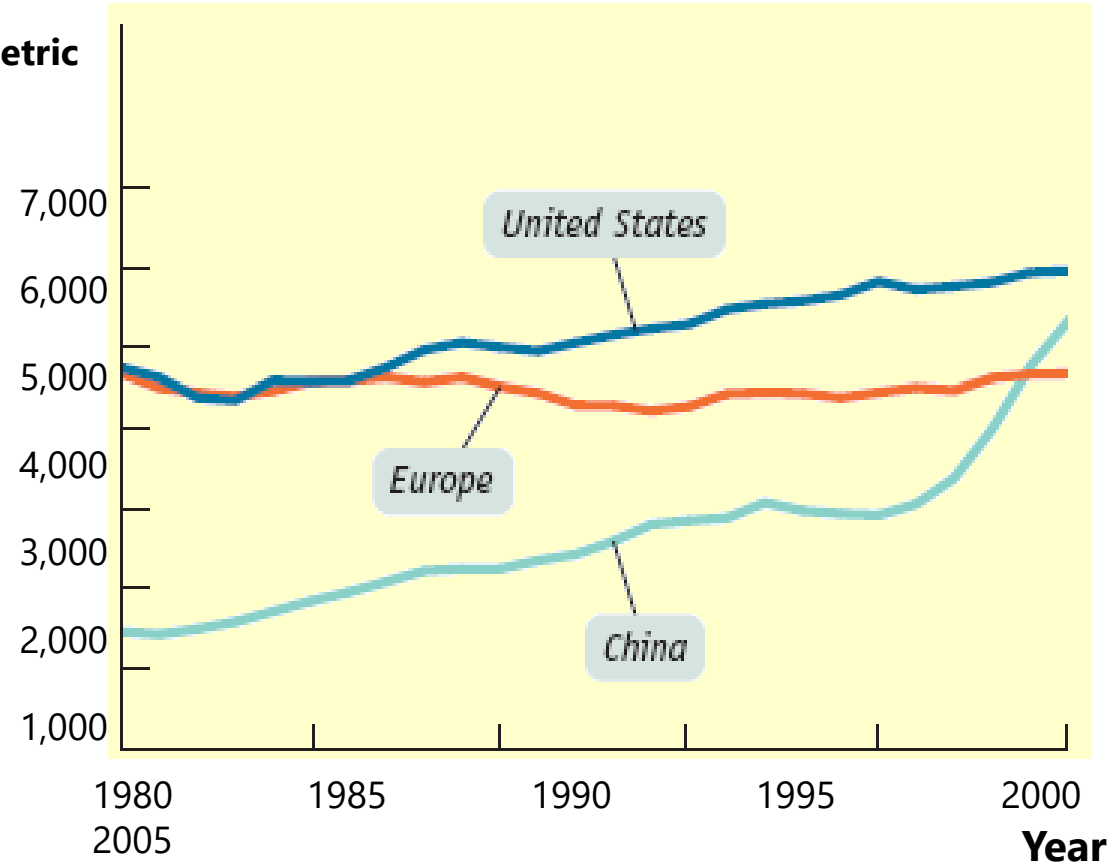
- Long-run economic growth is **sustainable** if it can continue in the face of the limited supply of natural resources and the impact of growth on the environment.
- Differing views about the impact of limited natural resources on long-run economic growth turn on the answers to three questions:
 - How large are the supplies of key natural resources?
 - How effective will technology be at finding alternatives to natural resources?
 - Can long-run economic growth continue in the face of resource scarcity?

Economic Growth and the Environment

- The limits to growth arising from environmental degradation are more difficult to overcome because overcoming them requires effective government intervention.
- The emission of greenhouse gases is clearly linked to growth, and limiting them will require some reduction in growth.
- However, the best available estimates suggest that a large reduction in emissions would require only a modest reduction in the growth rate.

Climate Change and Growth

Carbon dioxide
emissions
(million metric
tons)



Economic Growth and the Environment

- There is broad consensus that government action to address climate change and greenhouse gases should be in the form of market-based incentives, like a carbon tax or a cap and trade system.
- It will also require rich and poor countries to come to some agreement on how the cost of emissions reductions will be shared.

1. SUMMARY

2. Growth is measured as changes in real GDP per capita in order to eliminate the effects of changes in the price level and changes in population size. Levels of real GDP per capita vary greatly around the world: more than half of the world's population lives in countries that are still poorer than the United States was in 1907. Over the course of the twentieth century, real GDP per capita in the United States increased fivefold.
3. Growth rates of real GDP per capita also vary widely. According to the **Rule of 70**, the number of years it takes for real GDP per capita to double is equal to 70 divided by the annual growth rate of real GDP per capita.

3. The key to long-run economic growth is rising **labor productivity**, or just **productivity**, which is output per worker. Increases in productivity arise from increases in **physical capital** per worker and **human capital** per worker as well as advances in **technology**. The **aggregate production function** shows how real GDP per worker depends on these three factors. Other things equal, there are **diminishing returns to physical capital**: holding human capital per worker and technology fixed, each successive addition to physical capital per worker yields a smaller increase in productivity than the one before. **Growth accounting**, which estimates the contribution of each factor to a country's economic growth, has shown that rising **total factor productivity** is key to long-run growth. It is usually interpreted as the effect of technological progress.

4. The large differences in countries' growth rates are largely due to differences in their rates of accumulation of physical and human capital as well as differences in technological progress. A prime factor is differences in savings and investment rates. Technological progress is largely a result of **research and development**, or **R&D**.
5. Government actions that help growth are the building of **infrastructure**, particularly for public health, the creation and regulation of a well-functioning banking system that channels savings and investment spending, and the financing of both education and R&D. Government actions that retard growth are political instability, the neglect or violation of property rights, corruption, and excessive government intervention.

6. The world economy contains examples of success and failure in the effort to achieve long-run economic growth. East Asian economies have done many things right and achieved very high growth rates. In Latin America, where some important conditions are lacking, growth has generally been disappointing. In Africa, real GDP per capita has declined for several decades, although there are recent signs of progress. The growth rates of economically advanced countries have converged, but not the growth rates of countries across the world. This has led economists to believe that the **convergence hypothesis** fits the data only when factors that affect growth, such as education, infrastructure, and favorable policies and institutions, are held equal across countries.

7. Economists generally believe that environmental degradation poses a greater problem for whether long-run economic growth is **sustainable** than natural resource scarcity. Addressing environmental degradation requires effective governmental intervention, but the problem of natural resource scarcity is often well handled by the market price response.
8. The emission of greenhouse gases is clearly linked to growth, and limiting them will require some reduction in growth. However, the best available estimates suggest that a large reduction in emissions would require only a modest reduction in the growth rate.

9. There is broad consensus that government action to address climate change and greenhouse gases should be in the form of market-based incentives, like a carbon tax or a cap and trade system. It will also require rich and poor countries to come to some agreement on.

10. <https://www.youtube.com/watch?v=rqEtikPNo9M>

Exercises

- Answer Questions 10-12, Chapter 12, p.373
- Answer Questions 1-3, Chapter 12, p.375